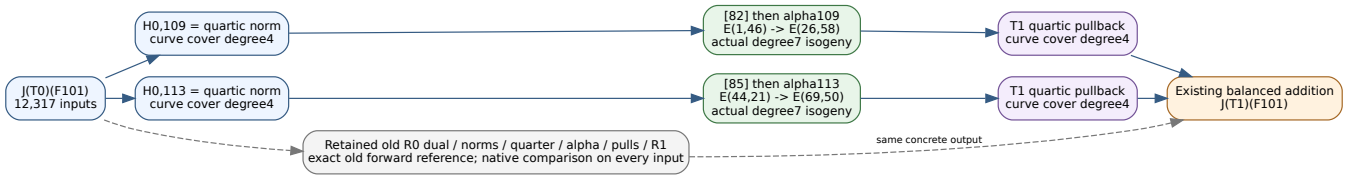
The first Jacobian seven-transfer using F101 arithmetic

2026-11-12 · fixed T0→T1 forward evaluator · scientific standing unsigned

The existing forward Psi now has a constructor and evaluator that use F101 arithmetic throughout. Native prototype15/PCC72e1d8 matches every 12,317 concrete output of the old transfer,with bijection,all fixed-partner sums,negations,identity and wrong-model controls. Selected alpha numerator, denominator,kernel and curve coefficients match the old maps exactly. This is an evaluator of the existing degree49 Jacobian map. [pcc:unverified]



Exact stages and their acceptance

The source and destination are the actual nonsplit sextics T0=[31,75,74,54,71,55,1] and T1=[95,26,19,27,94,91,1]. Both rational Jacobian groups have order12,317=109·113. The constructor takes accepted fixed covers and the selected elliptic kernel cubics[72,100,35,1] and [59,46,8,1]. It reuses existing Kohel and prime-field engines;parameter discovery remains inherited input provenance. [pcc:unverified]

Stage	Existing-engine evidence
Both T0 norms H0	Every12,317 old output/group/exception controls;585f35/2fcf0b
Quarter and alpha	Public inverse4 scalars82/85;coefficient-identical selected degree7 maps
T1 pullbacks K1	All109/113 inputs,old RiPi,norm4/off-diagonal0;5143f5
Balanced assembly	Original reduction arithmetic body byte-exact after extraction

All four T0/T1 pullbacks additionally pass49,300 ordered elliptic-group additions. Actual infinity multiplicities are[1,2] for T0 and[2,1] for T1. The two T1 covers each have116 finite-valued witnesses and two rational pole witnesses;all four geometric pole branches are retained. Separate flat-field replays5ab2bd/5eb5b1/850e92 validate the exact equations, degrees,complete source domains,infinities and poles. [pcc:unverified]

Why the replacement computes the same transfer

The accepted global factorization is4Psi=K1 alpha H0. Each H0 component is the pushforward of its exact degree4 cover;each K1 component is the corresponding T1 pullback. On the actual elliptic rational groups, 4·82=1 modulo109 and4·85=1 modulo113. Moving these public quarter scalars before alpha therefore gives Psi on J(T0)(F101),since4 is invertible on the order12,317 destination group. This uses no discrete-log labels and does not propose a global inverse morphism of[4]. The old adjoint norm7 preserves the original109/113 DLP components and recovery. PRIME_FIELD_PSI_PROOF.md/78176c records the complete argument. [pcc:unverified]

The curve maps are identified globally using exact polynomial equations, degree4 and a proved difference-degree bound16. More than16 distinct agreement points force equality. Old quotient/Picard/polarization/norm facts and expected-value provenance remain explicit dependencies. A negative ordinate passes the curve equation but fails the witnesses; equation fitting alone would not identify the map. [pcc:unverified]

Pullback jets and exact infinity bookkeeping

For finite P=(X,Y),form U=N-XQ²,V=YQ³/M moduloU. In the finite support algebra,F M²=Y²Q⁶. Supplied units Yi,Qi give the explicit inverse M inverse=F M Yi² Qi⁶,even if the quotient has nilpotents. Two bound QuarticFibre lemmas d4010b/4286c9 prove the actual cleared factorization and inverse,local kernel/axiom gate3413a4:2theorems,zero trusted native. Current136written helpers/25modules remain partial formalization. [pcc:unverified]

Since divQ=f*(O)-2(infinity++infinity-),a generic degree4 fibre has balanced plus count-1. At P=+q,a degree dropk gives plus=k-1;at P=-q it gives plus=-1. The Y sign is essential. The implemented rational elliptic groups have odd order;geometric nonzero2-torsion needs another chart and is outside this implemented domain. QUARTIC_PULLBACK_PROOF.md/ 4934b4 supplies the divisor derivation and original-engine custody. [pcc:unverified]

Actual integration and remaining validation

The new actual consumer child from6cf/ready1505 selects the prime-field entry in SevenChain.forward. Its exact old body remains forward_reference; the full comparison calls that explicit reference. Original dual,naive, source,geometric and chained controls remain. Integration02/1da990 passes strictClippy/source guards;complete changed guard03 is running on eight frozen sources,expected25native/90reference unit+five integrations,one pre-existing ignored benchmark. No completion is inferred here. [pcc:unverified]

Root standalone17 passes17tests/Clippyaf4b99. First-component consumer 1505 at6cf has complete25/85+five301fbe and separate own CI;published root59d2 CI99cdd2 covers15/134written/24modules,excluding later work. Unused-import failure7451f1 and intermediate19b3bd remain immutable. Independent whole-isogeny evaluation/kernel membership remains open. Matched cold construction,transfer,solve,recovery and verification costs await S3/profile/livebudget admission;no timing gain is claimed. [pcc:unverified]